

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T17 · Algebraic Roots & Indices

With Answers

Questions with full worked solutions

■ 88 pages

FAMILY MAP

Algebraic Roots Indices

F01

Combine indexed algebraic factors

24 Q | 48 marks

F02

Interpret zero, negative and fractional indices

11 Q | 23 marks

F03

Solve an index relation by a common base

7 Q | 18 marks

Working Unit 02

4MA1-1H Q4

ORIGINAL QUESTION**2 marks**

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$$m = \dots\dots\dots$$

(1)

$y^8 \div y^3 = y^n$

(b) Find the value of n

$$n = \dots\dots\dots$$

(1)

Worked Solution 02

Index laws: multiply and divide

UNIT 1
2 marks**Multiply equal bases**

$$x^5 x^7 = x^{5+7} = x^{12} \Rightarrow m = 12.$$

Divide equal bases

$$y^8 \div y^3 = y^{8-3} = y^5 \Rightarrow n = 5.$$

Final answer

$$m = 12, \quad n = 5$$

Working Unit 03

4MA1-1H Q4

ORIGINAL QUESTION

2 marks

(c) Simplify fully $(5a^4r^2)^3$

.....
(2)

Worked Solution 03

Power of a product

UNIT 1

2 marks

Apply the outer power

$$(5a^4r^2)^3 = 5^3(a^4)^3(r^2)^3 = 125a^{12}r^6.$$

Final answer

$$125a^{12}r^6$$

Working Unit 21

4MA1-2H Q15

ORIGINAL QUESTION

3 marks

15 Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

(Total for Question 15 is 3 marks)

Worked Solution 21

Negative power of an algebraic fraction

UNIT 1
3 marks**Simplify inside**

$$\frac{2a^3}{10a^7c^2} = \frac{1}{5a^4c^2}.$$

Use the negative power

$$\left(\frac{1}{5a^4c^2}\right)^{-3} = (5a^4c^2)^3 = 125a^{12}c^6.$$

Final answer

$$125a^{12}c^6$$

Question 06

4MA1-1H Q10

ORIGINAL QUESTION

4 marks

10 (a) Simplify $(2p)^0$ where $p > 0$

.....
(1)

$$y^9 \times y^{-3} = y^n$$

(b) Find the value of n

$n =$
(1)

(c) Simplify fully $(5a^4c^2)^3$

.....
(2)

Worked Solution 06

Index laws

MODULAR UNIT 1

4 marks

Zero power

$$(2p)^0 = 1.$$

Product

$$y^9 y^{-3} = y^6 \Rightarrow n = 6.$$

Power of a product

$$(5a^4 c^2)^3 = 125a^{12} c^6.$$

Final answer

$$1; \quad n = 6; \quad 125a^{12} c^6$$

Question 27

4MA1-2H Q23

ORIGINAL QUESTION

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

Worked Solution 27

Surds and index laws

MODULAR UNIT 1

3 marks

Rewrite in base 5

$$30 = 6 \cdot 5, 25^{2x+7} = 5^{4x+14}, \sqrt{180} = 6\sqrt{5}.$$

Combine

$$\frac{30 \cdot 25^{2x+7}}{\sqrt{180}(\sqrt{5})^{4x+9}} = 5^{2x+10}.$$

Final answer

$$5^{2x+10}$$

Question 17

4MA1-2H Q4 (b)

ORIGINAL QUESTION UNIT 1

1 marks

(b) Simplify $h^{15} \div h^3$
(1)

Worked solution

METHOD / WORKING

MODULAR UNIT 1 • NOVEMBER 2023 DETAILED SOLUTIONS
Worked solution 17 MODULAR UNIT 1
Index law for division 1 marks
(b) – Index law for division
Subtract the powers
$h^{15} \div h^3 = h^{15-3} = h^{12}$.
Final answer
h^{12}

FINAL ANSWER

Question 18

4MA1-2H Q4 (c)

ORIGINAL QUESTION UNIT 1

2 marks

(c) Simplify fully $(2g^3k^5)^4$

.....
(2)

Worked solution

METHOD / WORKING

MODULAR UNIT 1 • NOVEMBER 2023 DETAILED SOLUTIONS
Worked solution 18 MODULAR UNIT 1
Power of a product 2 marks
(c) – Power of a product
Raise each factor to the fourth power
$(2g^3k^5)^4 = 2^4g^{3 \cdot 4}k^{5 \cdot 4}$.
Simplify
$(2g^3k^5)^4 = 16g^{12}k^{20}$.
Final answer
$16g^{12}k^{20}$

FINAL ANSWER

Question 19

4MA1-2H Q4 (d)

ORIGINAL QUESTION UNIT 1

2 marks

- (d) Given that $\frac{y^5 \times y^n}{y^7} = y^{12}$
work out the value of n

$n = \dots\dots\dots$
(2)

Worked solution 19

Find an unknown index

MODULAR UNIT 1

2 marks

(d) – Find an unknown index

Combine powers

$$\frac{y^5 y^n}{y^7} = y^{5+n-7} = y^{n-2}.$$

Equate exponents

$$n - 2 = 12 \Rightarrow n = 14.$$

Final answer

$$n = 14$$

Question 25

4MA1-2H Q12 (b)

ORIGINAL QUESTION UNIT 1

2 marks

(b) Simplify fully $(125p^6y^{24})^{\frac{2}{3}}$

.....
(2)

Worked solution

METHOD / WORKING

MODULAR UNIT 1 • NOVEMBER 2023 DETAILED SOLUTIONS
Worked solution 25 MODULAR UNIT 1
Fractional power of a product 2 marks
(b) – Fractional power of a product
Apply the two-thirds power
$(125p^6y^{24})^{2/3} = 125^{2/3}p^4y^{16}$.
Simplify the coefficient
$125^{2/3} = (5^3)^{2/3} = 5^2 = 25$, hence $25p^4y^{16}$.
Final answer
$25p^4y^{16}$

FINAL ANSWER

Question 02

4MA1-1H Q6

ORIGINAL QUESTION

2 marks

6 (a) Simplify $(2c^4d^7)^3$

(2)

Worked Solution 02

Power of a monomial

MODULAR UNIT 1

2 marks

Apply the outer power

$$(2c^4d^7)^3 = 2^3c^{12}d^{21}.$$

Final answer

$$8c^{12}d^{21}$$

Question 03

4MA1-1H Q6

ORIGINAL QUESTION

1 marks

(b) Find the value of $5y^0$ where $y > 0$

(1)

Worked Solution 03

Zero index

MODULAR UNIT 1

1 marks

Use the zero-power rule

$$y^0 = 1 \quad (y > 0), \quad 5y^0 = 5.$$

Final answer

5

Question 19

4MA1-2H Q15

ORIGINAL QUESTION

2 marks

15 (a) Simplify fully $(32a^{15})^{\frac{3}{5}}$

(2)

Worked Solution 19

Fractional index power

MODULAR UNIT 1

2 marks

Apply the outer power

$$(32a^{15})^{3/5} = 32^{3/5}a^9 = (2^5)^{3/5}a^9 = 8a^9.$$

Final answer

$$8a^9$$

Question 20

4MA1-2H Q15

ORIGINAL QUESTION

2 marks

(b) Express $\left(\frac{1}{10x}\right)^{-3}$ in the form px^n where p and n are integers.

.....
(2)

Worked Solution 20

Negative index power

MODULAR UNIT 1

2 marks

Invert the bracket

$$\left(\frac{1}{10x}\right)^{-3} = (10x)^3.$$

Expand

$$(10x)^3 = 1000x^3.$$

Final answer

$$1000x^3$$

Original question 01

4MA1-1H Q3 M01

MODULAR UNIT 1

1 marks

3 (a) Simplify $(p^3)^5$

(1)

Worked solution 01

Simplify a power of a power

MODULAR UNIT 1

1 marks

M01 – Simplify a power of a power**Use the power rule**

$$(p^3)^5 = p^{3 \times 5} = p^{15}.$$

Final answer

$$p^{15}$$

Original question 23

4MA1-2H Q14 M01

MODULAR UNIT 1

3 marks

14 Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n

Show your working clearly.

$n = \dots\dots\dots$

Worked solution 23

Solve an index equation

MODULAR UNIT 1

3 marks

M01 – Solve an index equation**Combine powers of 3**

The left side has exponent $2n-1$; the right side has exponent $4-2n$.

Equate exponents

$$2n - 1 = 4 - 2n.$$

Solve

$$4n = 5 \Rightarrow n = 5/4.$$

Final answer

$$n = \frac{5}{4}$$

Original question 04

4MA1-1H Q9 Q9(a)

ORIGINAL QUESTION**1 marks**

9 (a) Simplify $8 \times (4t)^0$

(1)

Worked solution 04

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

1 marks

Q9(a) – Chapter: Algebraic Roots & Indices

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

MethodSince $(4t)^0 = 1$, $8 \times (4t)^0 = 8$.*Why: Any nonzero base raised to the power zero equals 1.***Final answer**

simplified value: 8

Original question 05

4MA1-1H Q9 Q9(b)

ORIGINAL QUESTION**1 marks**

$$x^6 \div x^{-5} = x^p$$

(b) Find the value of p $p = \dots\dots\dots$
(1)

Worked solution

Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 05 MODULAR UNIT 1
Chapter: Algebraic Roots & Indices 1 marks
Q9(b) – Chapter: Algebraic Roots & Indices
Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases
Method
$x^6/x^{-5} = x^{(6 - (-5))} = x^{11}, \text{ so } p = 11.$
Why: When dividing like bases, subtract the denominator exponent.
Final answer
p: $p = 11$

FINAL ANSWER

Original question 06

4MA1-1H Q9 Q9(c)

ORIGINAL QUESTION**2 marks**(c) Simplify fully $(2k^2m^4)^3$

(2)

Worked solution

Family: Combine indexed algebraic factors • Variant: Power of a product or power

METHOD / WORKING

MODULAR UNIT 1 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 06 MODULAR UNIT 1
Chapter: Algebraic Roots & Indices 2 marks
Q9(c) – Chapter: Algebraic Roots & Indices
Family: Combine indexed algebraic factors • Variant: Power of a product or power
Method
$(2k^2m^4)^3 = 2^3(k^2)^3(m^4)^3.$
Why: Apply the outside power to every factor .
Method
<i>This simplifies to $8k^6m^{12}$.</i>
Why: Multiply the nested exponents and evaluate 2^3 .
Final answer
<i>simplified expression : $8k^6m^{12}$</i>

FINAL ANSWER

Original question 29

4MA1-2H Q24 Q24

ORIGINAL QUESTION**3 marks****24**

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

Worked solution 29

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

3 marks

Q24 – Chapter: Algebraic Roots & Indices

Family: Solve an index relation by a common base • Variant: Express one exponent variable in terms of others

Method

Use $18 = 2 \times 3^2$, $\sqrt{27} = 3^{(3/2)}$, $6 = 2 \times 3$, and $9 = 3^2$.

Why: Rewrite every factor using base 3 where possible.

Method

The left side becomes $[2 \times 3^2 \times 3^{(6n+9)}] / [2 \times 3 \times 3^{(4n+16)}]$.

Why: Multiply the nested exponents and collect the fixed factors.

Method

This simplifies to $3^{(6n+11-4n-17)} = 3^{(2n-6)}$, so $x = 2n - 6$.

Why: Subtract denominator exponents and equate equal bases.

Final answer

x: $x = 2n - 6$

Original question 01

4MA1-1H Q1 Q1(a)

ORIGINAL QUESTION

1 marks

1 (a) Simplify $a^7 \times a^4$

.....
(1)

Worked solution

Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 01 MODULAR UNIT 1
Chapter: Algebraic Roots & Indices 1 marks
Q1(a) – Chapter: Algebraic Roots & Indices
Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases
Method
$a^7 \times a^4 = a^{(7+4)} = a^{11}$.
Why: When multiplying powers with the same base, add the indices.
Final answer
<i>simplified form</i> : a^{11}

FINAL ANSWER

Original question 02

4MA1-1H Q1 Q1(b)

ORIGINAL QUESTION**1 marks**(b) Simplify $w^{15} \div w^3$

(1)

Worked solution

Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 02 MODULAR UNIT 1
Chapter: Algebraic Roots & Indices 1 marks
Q1(b) – Chapter: Algebraic Roots & Indices
Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases
Method
$w^{15} \div w^3 = w^{(15 - 3)} = w^{12}$.
Why: When dividing powers with the same base, subtract the indices.
Final answer
<i>simplified form</i> : w^{12}

FINAL ANSWER

Original question 03

4MA1-1H Q1 Q1(c)

ORIGINAL QUESTION**2 marks**(c) Simplify $(8x^5y^3)^2$

(2)

Worked solution

Family: Combine indexed algebraic factors • Variant: Power of a product or power

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 03 MODULAR UNIT 1
Chapter: Algebraic Roots & Indices 2 marks
Q1(c) – Chapter: Algebraic Roots & Indices
Family: Combine indexed algebraic factors • Variant: Power of a product or power
Method
$(8x^5y^3)^2 = 8^2x^{10}y^6$.
Why: Square the coefficient and multiply each exponent by 2.
Method
<i>Therefore</i> $(8x^5y^3)^2 = 64x^{10}y^6$.
Why: $8^2 = 64$.
Final answer
<i>simplified form</i> : $64x^{10}y^6$

FINAL ANSWER

Original question 11

4MA1-1H Q13 Q13(b)

ORIGINAL QUESTION**3 marks**

(b) Simplify completely $\left(\frac{16w^8}{y^{20}}\right)^{-\frac{3}{4}}$

(3)

Worked solution

Family: Combine indexed algebraic factors • Variant: Complex indexed monomial or fractional outer

METHOD / WORKING

MODULAR UNIT 1 • JANUARY 2022 DETAILED SOLUTIONS
Worked solution 11 MODULAR UNIT 1
Chapter: Algebraic Roots & Indices 3 marks
Q13(b) – Chapter: Algebraic Roots & Indices
Family: Combine indexed algebraic factors • Variant: Complex indexed monomial or fractional outer
power
Method
$(16w^8/y^{20})^{(-3/4)} = y^{15}/(16^{(3/4)}w^6).$
Why: Apply the $-3/4$ power to each factor and invert.
Method
$16^{(3/4)} = 8 \text{ and } w^{(-6)} = 1/w^6.$
Why: Evaluate the coefficient and rewrite the negative exponent.
Method
The simplified result is $y^{15}/(8w^6).$
Why: Combine the factors.
Final answer
simplified expression : $y^{15}/(8w^6)$

FINAL ANSWER

(c) Simplify $(p^2 + 3)^0$

(1)

Worked solution

January 2020 | Question 7 | 1 marks

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Method / working

1. Any non-zero expression to the zero power is 1.

Final answer: 1

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

(3)

Worked solution

January 2020 | Question 17 | 3 marks

Family: Combine indexed algebraic factors | Variant: Complex indexed monomial or fractional outer power

METHOD / WORKING

1. *Invert the negative power*: $(t^{15}/(27a^{12}))^{(2/3)}$.
2. Apply the power to numerator and denominator.
3. *Simplify* to $t^{10}/(9a^8)$.

FINAL ANSWER

$$t^{10}/(9a^8)$$

1 (a) Simplify $\frac{x^9}{x^2}$

(1)

Worked solution

January 2020 | Question 1 | 1 marks

Family: Combine indexed algebraic factors | Variant: Multiply or divide common bases

METHOD / WORKING

$$1. x^9/x^2 = x^{(9-2)} = x^7.$$

FINAL ANSWER

$$x^7$$

(b) Write $\frac{7^8 \times 7^4}{7^3}$ as a single power of 7

(2)

Worked solution

January 2020 | Question 1 | 2 marks

Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases

Method / working

1. For common base 7, add 8+4 then subtract 3.
2. $7^8 \times 7^4 \div 7^3 = 7^9$.

Final answer: 7^9

9 (a) Simplify $(2x^3y^5)^4$

(2)

Worked solution

January 2021 | Question 9 | 2 marks

Family: Combine indexed algebraic factors | Variant: Power of a product or power

METHOD / WORKING

1. *Raise each factor* to the fourth power: $2x^1y^2$.

2. *So the result is* $16x^{12}y^2$.

FINAL ANSWER

$$16x^{12}y^2$$

Given that $\frac{y^5 \times y^n}{y^6} = y^{13}$

(b) work out the value of n .

$n = \dots\dots\dots$
(2)

Worked solution

January 2021 | Question 5 | 2 marks

Family: Solve an index relation by a common base • Variant: Find a numerical unknown exponent

Method / working

1. Use index laws: $y^a \times y^b / y^c = y^{a+b-c}$.

2. Set $n-1=13$, so $n=14$.

Final answer: $n=14$

6 (a) Write down the value of y^0

(1)

Worked solution

January 2021 | Question 6 | 1 marks

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Method / working

1. Any non-zero base to the zero power equals 1.

Final answer: 1

11 Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}} \right)^{-\frac{1}{2}}$

Worked solution

January 2021 | Question 11 | 3 marks

Family: Combine indexed algebraic factors | Variant: Complex indexed monomial or fractional outer power

METHOD / WORKING

1. Cancel common numerical, t and w factors first.
2. Apply the outside negative power and square to the remaining bracket.
3. The simplified result is $4t^{-2}w^2$.

FINAL ANSWER

$$4t^{-2}w^2$$

Given that $\frac{w^5 \times w^n}{w^3} = w^{10}$

(c) work out the value of n .

$n = \dots\dots\dots$
(2)

Worked solution

June 2021 | Question 5 | 2 marks

Family: Solve an index relation by a common base | Variant: Find a numerical unknown exponent

METHOD / WORKING

1. Simplify the printed relation $(w \times w) / w^3 = w^1$ to $w^2 = w^1$ (equivalently multiply through to get $w \times w = w^1 \times 3$).
2. Equate powers: $n+2=10$ (or $5+n=13$), giving $n=8$.

FINAL ANSWER

 $n = 8$

11 $\sqrt{2} \times 16 = 2^x$

- (a) Find the value of x .
Show your working clearly.

$x = \dots\dots\dots$
(2)

Worked solution

June 2021 | Question 11 | 2 marks

Family: Interpret zero, negative and fractional indices • Variant: Fractional indices as algebraic roots

Method / working

1. Rewrite the powers as powers of 2 and equate exponents.
2. Solve $2x=9$ to obtain $x=9/2=4.5$.

Final answer: $x = 9/2$

$$\frac{(11^{-6})^5}{11^4} = 11^n$$

- (b) Find the value of n .
Show your working clearly.

$n = \dots\dots\dots$
(2)

Worked solution

June 2021 | Question 11 | 2 marks

Family: Solve an index relation by a common base • Variant: Find a numerical unknown exponent

Method / working

1. The exponent in $(11^{\frac{1}{5}})^n$ is $(-6) \times 5 = -30$, then division by $11^{\frac{1}{5}}$ gives exponent -34 .
2. Equate with $11^{\frac{1}{5}}$ to obtain $n = -34$.

Final answer: $n = -34$

14 Simplify fully $\left(\frac{9x^4}{16y^{10}}\right)^2$

Worked solution

June 2021 | Question 14 | 3 marks

Family: Combine indexed algebraic factors | Variant: Complex indexed monomial or fractional outer power

METHOD / WORKING

1. Apply the negative half power to the numerical fraction: $\sqrt[3]{16/9} = 4/3$.2. Apply the powers to x and y: $x^{-1/2} \rightarrow x^{-1/2 \times 2} = x^{-1}$ and $y^{1/3} \rightarrow y^{1/3}$.3. Combine: $4y^{1/3}/(3x^2)$.

FINAL ANSWER

$$4y^{1/3}/(3x^2)$$

8 (a) Simplify $(3x^2y)^0$

(1)

Worked solution

November 2021 | Question 8 | 1 marks

Family: Interpret zero, negative and fractional indices | Variant: Zero and negative indices

METHOD / WORKING

1. (a) $(3x^2y)^0 = 1.$

FINAL ANSWER

1, $(x-9)(x+4)$, $x=9$ or $x=-4$

12 (a) Simplify $(64p^9q^{12})^{\frac{2}{3}}$

(2)

Worked solution

November 2021 | Question 12 | 2 marks

Family: Combine indexed algebraic factors | Variant: Complex indexed monomial or fractional outer power

METHOD / WORKING

1. (a)

$$2. (64p^9q^{12})^{2/3} = 16p^6q^8$$

FINAL ANSWER

$$(a) 16p^6q^8$$

12 Given that $\frac{3^x}{9^{3x}} = 81$

find the value of x .

Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 12 is 3 marks)

Answer reference | May-Nov 2020 | Question 12

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

12	$3^4 = \frac{3^x}{9^{3x}}$ or $81 = \frac{3^x}{(3^2)^{3x}}$	$9^2 = \frac{3^x}{9^{3x}}$ or $81 = \frac{(9^{0.5})^x}{9^{3x}}$			M1 replacing 81 with 3^4 or 9^{3x} with $(3^2)^{3x}$ (or 3^{6x}) or replacing 81 with 9^2 or 3^x with $(9^{0.5})^x$ (in an equation)
	eg $4 + 6x = x$ or $4 = x - 2(3x)$ oe	eg $2 = 0.5x - 3x$ oe			M1 a correct equation using powers
			-0.8	3	A1 oe, dep on at least M1
					Total 3 marks

Worked solution

May-Nov 2020 | Question 12 | 3 marks

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Official final mark-scheme pages are embedded immediately after this question.

12 $3^{\frac{1}{2}} \times 3^{\frac{2}{5}} = 3^m$

(a) Work out the value of m

$m = \dots\dots\dots$
(1)

$5^{-10} \div 5^{-4} = 5^n$

(b) Work out the value of n

$n = \dots\dots\dots$
(1)

(Total for Question 12 is 2 marks)

Answer reference | January 2023 | Question 12

Family: Interpret zero, negative and fractional indices • Variant: Fractional indices as algebraic roots

12	(a)		$\frac{9}{10}$	1	B1 oe
	(b)		-6	1	B1
					Total 2 marks

Worked solution

January 2023 | Question 12 | 2 marks

Family: Interpret zero, negative and fractional indices • Variant: Fractional indices as algebraic roots

Official final mark-scheme pages are embedded immediately after this question.

26 Find the values of n such that

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

Show clear algebraic working.

(Total for Question 26 is 5 marks)

Answer reference | January 2023 | Question 26

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Q	working	Answer	Mark	Notes
26	For one of: $10^{4n} = (5 \times 2)^{4n}$ or $5^{4n} \times 2^{4n}$ or oe $20^2 = (5 \times 2^2)^2$ or $5^2 \times 2^4$ or $5^2 \times 2^2 \times 2^2$ or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4		5	M1 for writing 10^{4n} correctly as a product of 5 and 2 to the power $4n$ oe or writing 20^2 correctly as a product of 5 and 2^2 to the power 2 oe or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4
	for getting numerator to the stage (this scores M1M1) $2^{4n} \times 2^{3n^2-15n} \times 5^2$ oe eg $2^{3n^2-11n} \times 5^{-4n+4n+2}$ oe or for two of $10^{4n} = (5 \times 2)^{4n}$ or $5^{4n} \times 2^{4n}$ or oe $20^2 = (5 \times 2^2)^2$ or $5^2 \times 2^4$ or $5^2 \times 2^2 \times 2^2$ or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4 or getting the numerator to the stage			M1
	$\frac{2^{4n} \times 2^{3n^2} \times 2^{-15n}}{2^4} [= 1 \text{ or } 2^0]$ or $2^{4n} \times 2^{3n^2} \times 2^{-15n} = 2^4$ oe			M1 For writing the equation in powers of 2 only
	$3n^2 - 11n - 4 [= 0]$			A1 Correct quadratic equation dep on M1
	working required	$-\frac{1}{3}, 4$		A1 Both answers required dep on M1
Total 5 marks				

Worked solution

January 2023 | Question 26 | 5 marks

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Official final mark-scheme pages are embedded immediately after this question.

8 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n

$n = \dots\dots\dots$

(Total for Question 8 is 2 marks)

Worked solution

November 2025 | Question 8 | 2 marks

Family: Combine indexed algebraic factors | Variant: Multiply or divide common bases

METHOD / WORKING

Worked solution
November 2025 Question 8 2 marks
Family: Combine indexed algebraic factors Variant: Multiply or divide common bases
Solution
$\frac{8^{(-2)} \times 8^9}{8^{(10)}} = 8^{(-2 + 9 - 10)} = 8^{(-3)}$
Answer: $n=-3$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER

13 (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^{\frac{2}{3}}$

.....
(2)

Given that $4^n = \frac{4^m}{64^p}$

(b) Express n in terms of m and p

$n =$
(2)

(Total for Question 13 is 4 marks)

Worked solution

November 2025 | Question 13 | 4 marks

Family: Solve an index relation by a common base | Variant: Find a numerical unknown exponent

METHOD / WORKING

Worked solution
November 2025 Question 13 4 marks
Family: Solve an index relation by a common base Variant: Find a numerical unknown exponent
Solution
(a)
$(\frac{125x^6}{y^{15}})^{1/3}$
$= (5x^2)/(y^5)$
(b)
$64^p = (4^3)^p = 4^{3p}$
$4^n = (4^m)/(64^p) = \frac{4^m}{4^{3p}} = 4^{m-3p}$
Answer: (a) $(5x^2)/(y^5)$, (b) $n = m - 3p$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER